



**CSE422 : Artificial Intelligence
Project Report**

Project Title : E-commerce Shipping Delivery Prediction

Group No : 11, CSE422 Lab Section : 06, Spring 2026	
ID	Name
24101416	Afri Anjum Tutul
24101279	Dhrubojyoti Saha

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Introduction

This project aims to predict whether an e-commerce shipment will reach its destination on time based on various factors like the mode of shipment, customer ratings, cost of the product, weight, discounts and more.

Predicting on-time delivery can help the company improve the customer satisfaction and overall rating.

Dataset Description

After removing the non-informative ID column and separating the target variable, the dataset contains 10 input features and 1 target variable(Reached.on.Time_Y.N).

This is a supervised binary classification problem because the target variable contains 0 or 1. The dataset consists of approximately 10,000 datapoints. It has both quantitative

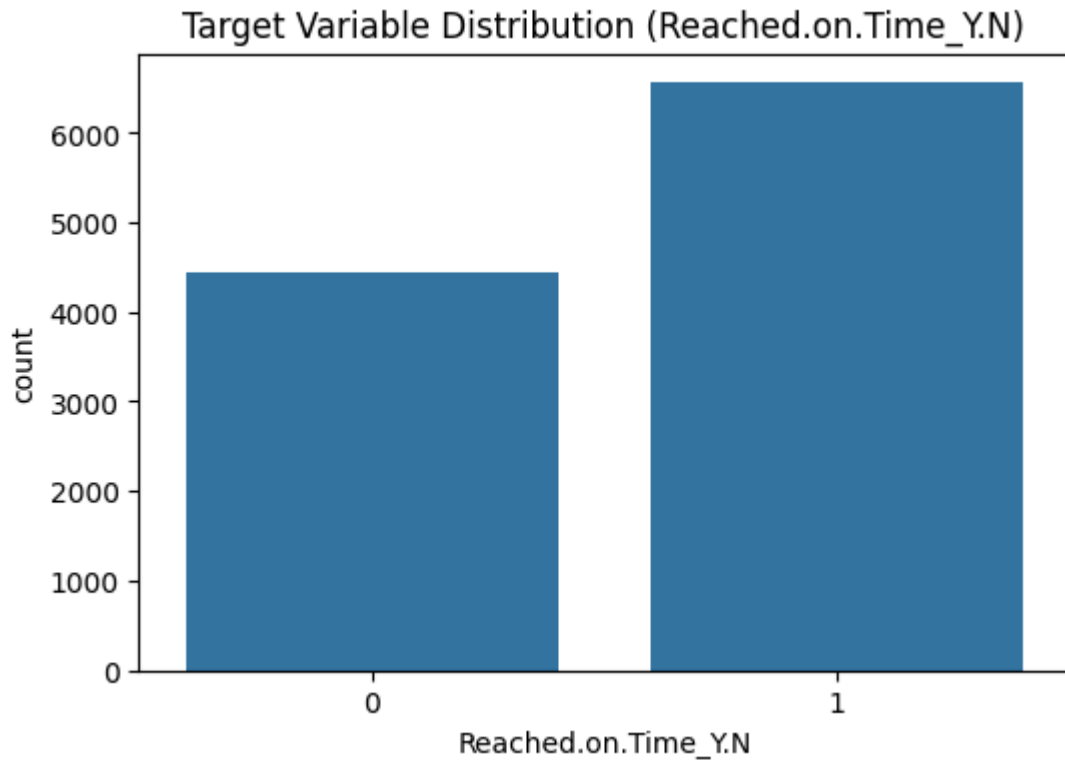
(Customer_care_calls, Customer_rating, Cost_of_the_Product, Prior_purchases, Discount_offered, Weight_in_gms) and categorical features (Warehouse_block, Mode_of_Shipment, Product_importance, Gender).

We need to encode the categorical feature into numerical as machine learning models require numerical input.

Imbalance Dataset

The target variable Reached.on.Time_Y.N contains two classes:

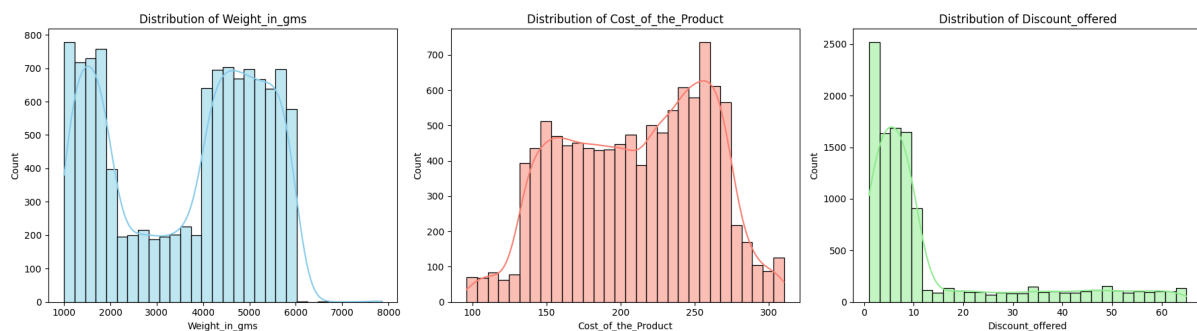
- 0 → Delivered on time
- 1 → Delivered late



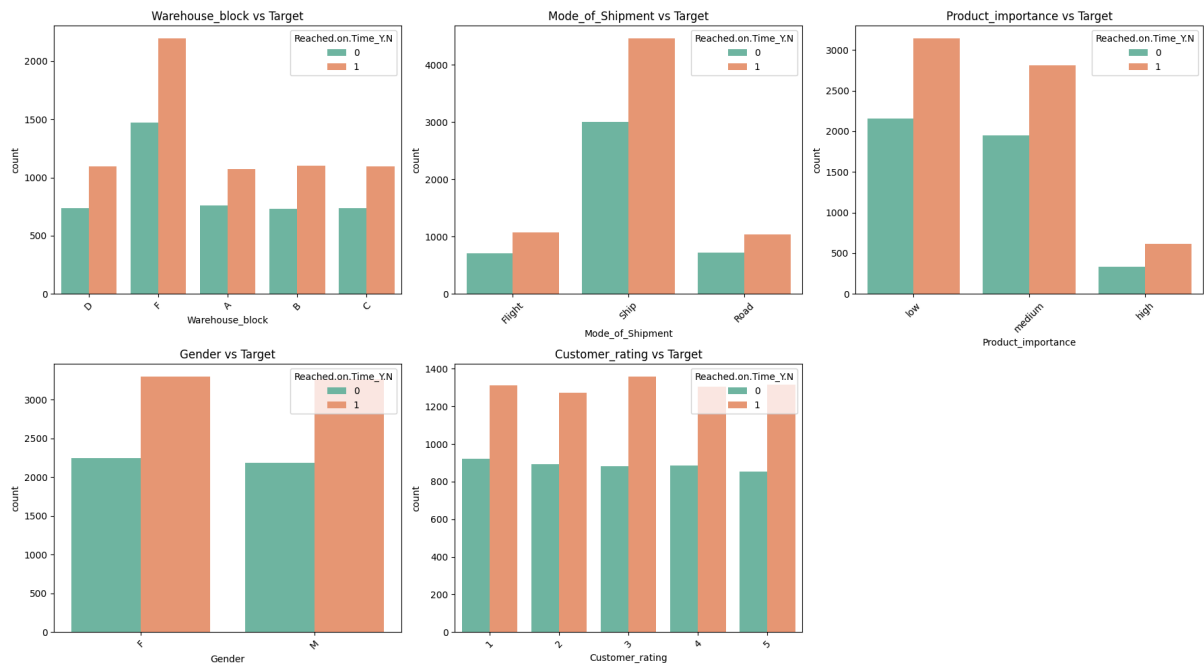
The dataset is relatively balanced, as both classes (on-time and delayed deliveries) have comparable numbers of instances. Therefore, no extreme resampling techniques such as oversampling or undersampling are required. However, appropriate evaluation metrics should still be considered.

EDA(Exploratory Data Analysis)

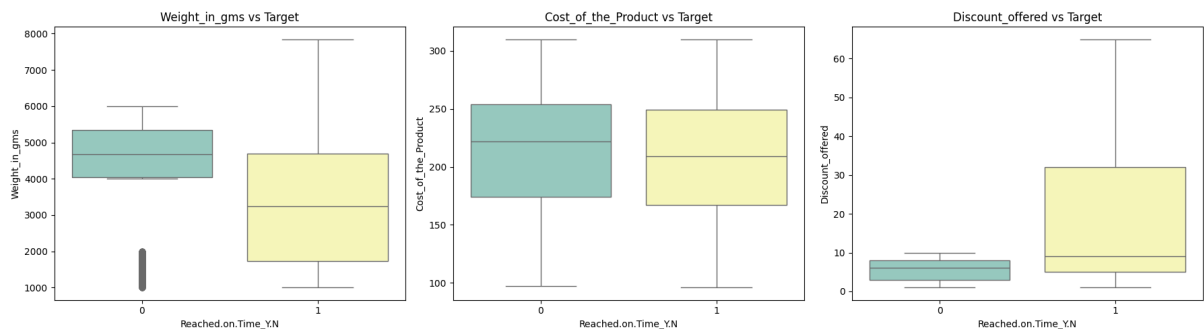
Through exploratory data analysis, many visualization methods were used to understand the features and their relationships.



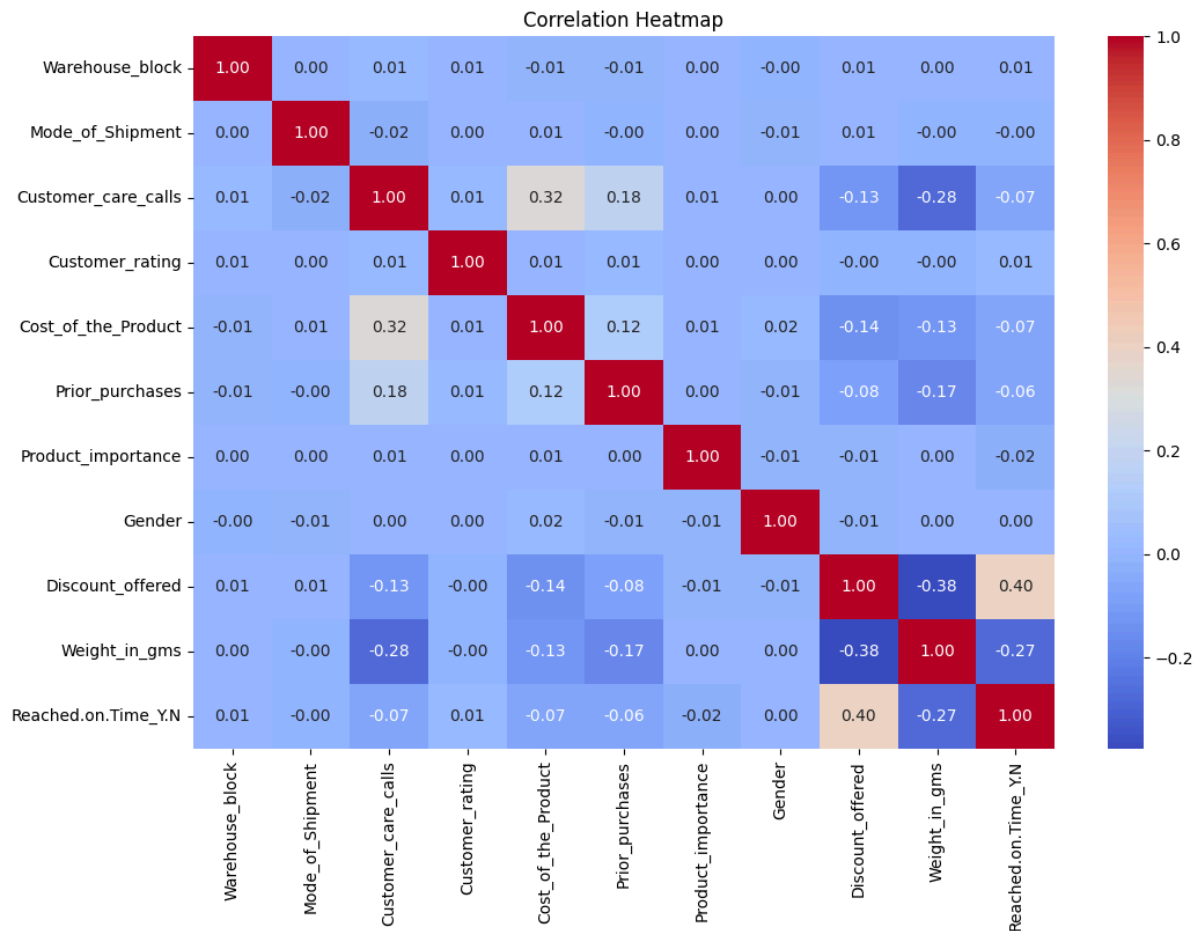
We used **Histogram** to show the varying distribution and scales of numerical features with some skewness and outliers, which suggested the need for preprocessing such as standard scaling.



Bar charts show that categorical variables like Warehouse_block and Mode_of_Shipment influence the outcome of delivery, indicating their predictive importance.



Moreover, the **Box plot** diagram shows that Discount_offered has notably higher values and significant outliers for delayed shipments, indicating it as a strong indicator of delivery delay.



Lastly, **correlation heatmap** was used to measure the linear relationships between numerical features and the target variable. The results show that **Discount_offered** has the highest positive correlation with delivery delay, indicating that higher discounts are related with a greater probability of late shipments. Whereas **Weight_in_gms** is seen to have a moderate negative correlation, which implies that if weight decreases, the chances of delay increases. Beside that, most other features show weak correlations, which suggests that the dataset does not have strong linear dependencies.

It is mathematically confirmed that 'Discount_offered' has the highest positive correlation with late deliveries.

Dataset pre-processing

In the data pre-processing , there are no missing values present in any features. So, no preprocessing steps were needed for handling null values.

The dataset contains categorical feature like Warehouse_block,Mode_of_Shipment,

Product_importance, Gender. As machine learning models cannot process categorical value directly we used level encoder.

Dataset Splitting

The dataset was divided into training and testing subsets using a stratified sampling technique. 80% of the data was allocated to the training set and 20% to the testing set. Stratification was applied based on the target variable to ensure that the proportion of classes remains consistent across both subsets. This helps prevent bias that could arise from uneven class distribution. The training set is used to build and learn the model, while the testing set is reserved for evaluating the model's performance on unseen data.

From histogram we observe that the numerical features have different ranges and distributions like Cost_of_the_Product is in smaller range and Weight_in_gms in very large range, which can negatively affect certain models. So Standard scaling was applied to normalise the numerical value range, ensuring that no feature dominates others due to scale differences.

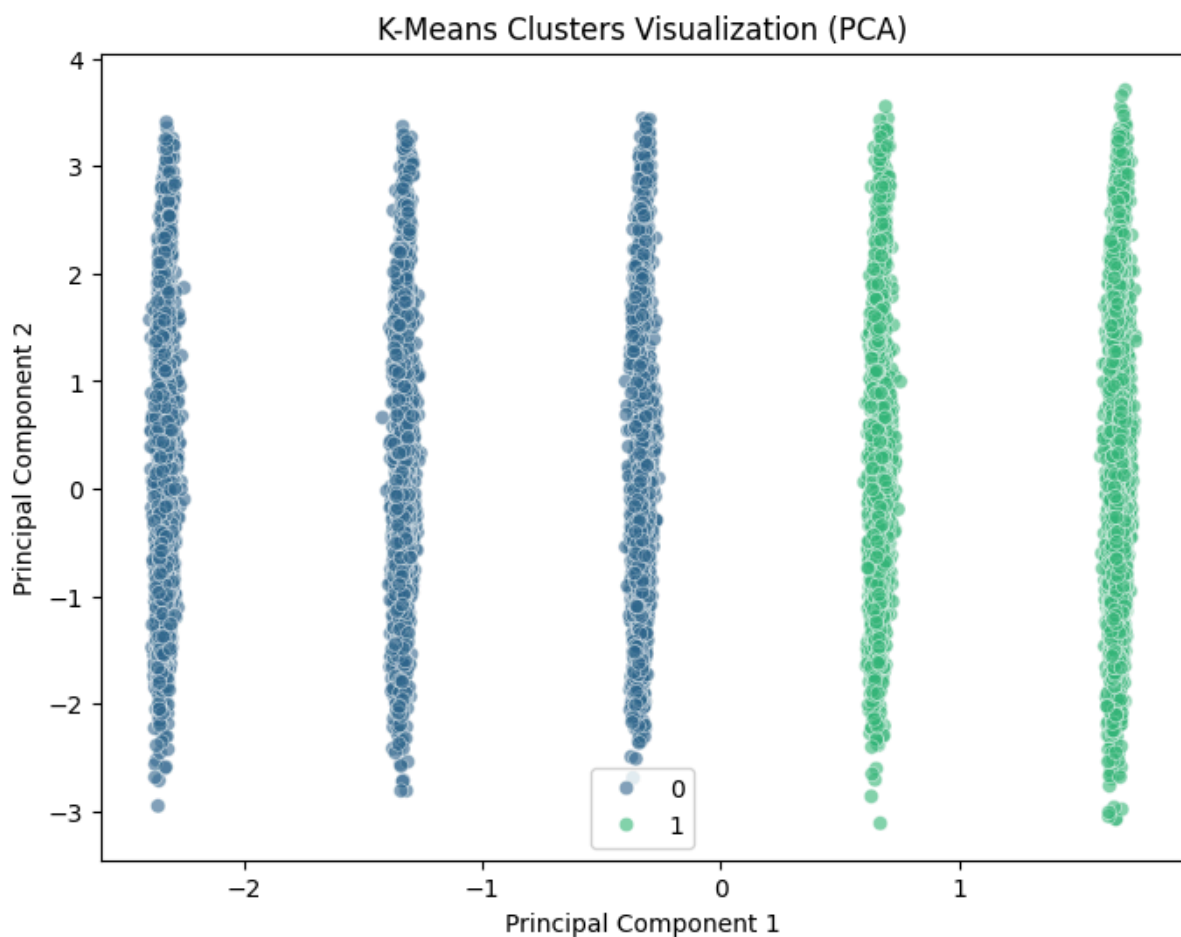
Model training and Testing

In this project, both supervised and unsupervised machine learning approaches were applied to analyze delivery delay prediction. For the supervised learning part, the dataset was divided into training and testing sets using an 80:20 split with stratification to preserve the class distribution. Models including Logistic Regression, Decision Tree, and Neural Network were trained on the training dataset (X_train, y_train), where they learned the relationship between input features and the target variable (Reached.on.Time_Y.N).

Logistic Regression captured linear relationships, Decision Tree learned rule-based patterns through feature splits, and the Neural Network modeled complex non-linear interactions.

The trained models were then evaluated on the unseen test dataset (X_test) by generating predictions and comparing them with actual labels (y_test), ensuring proper generalization and avoiding overfitting.

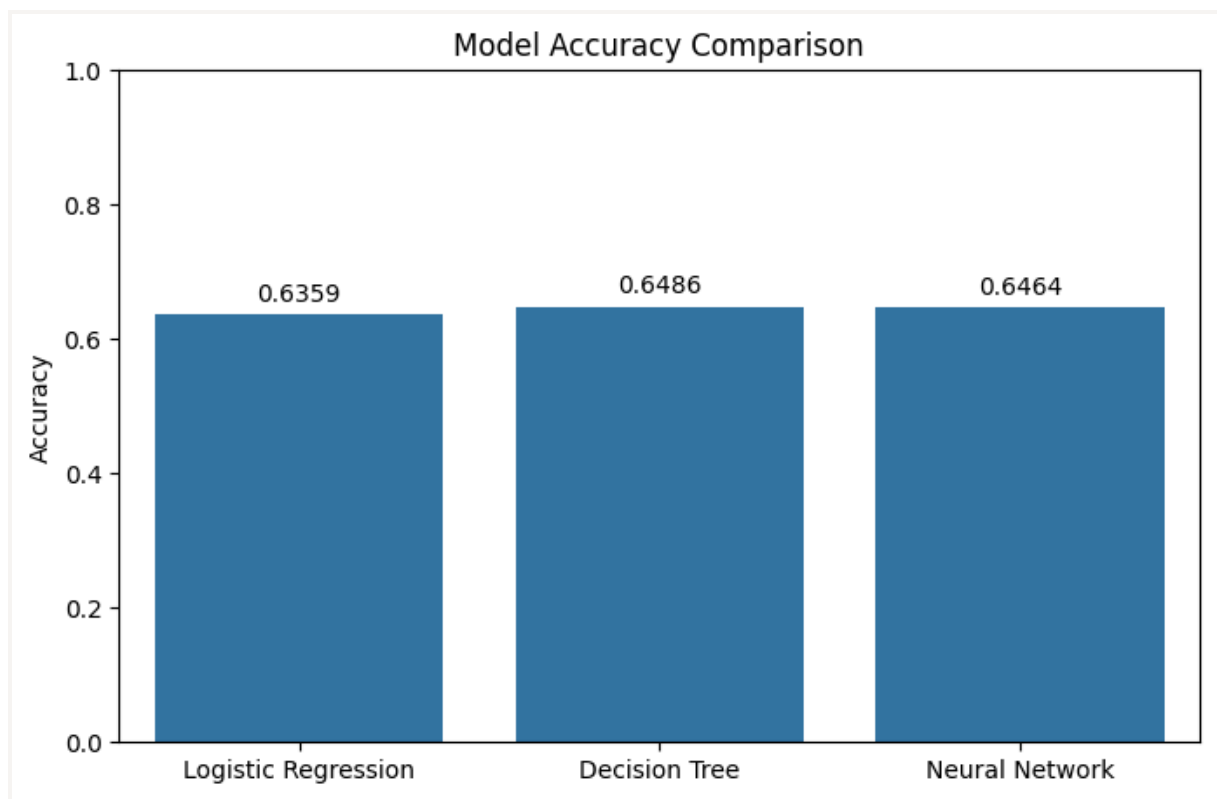
In addition to supervised learning, K-Means clustering was applied as an unsupervised learning technique by removing the target variable and using only the input features for training. Since K-Means does not use label information, the algorithm grouped the data into two clusters based purely on feature similarity and distance between data points. PCA was then applied to reduce the high-dimensional data into two principal components for visualization. The resulting cluster plot showed noticeable separation mainly along the first principal component, indicating that the dataset contains some underlying structure. However, the spread of points within each cluster also suggests internal variability and overlapping feature patterns, meaning the data is not perfectly separable through unsupervised clustering alone.



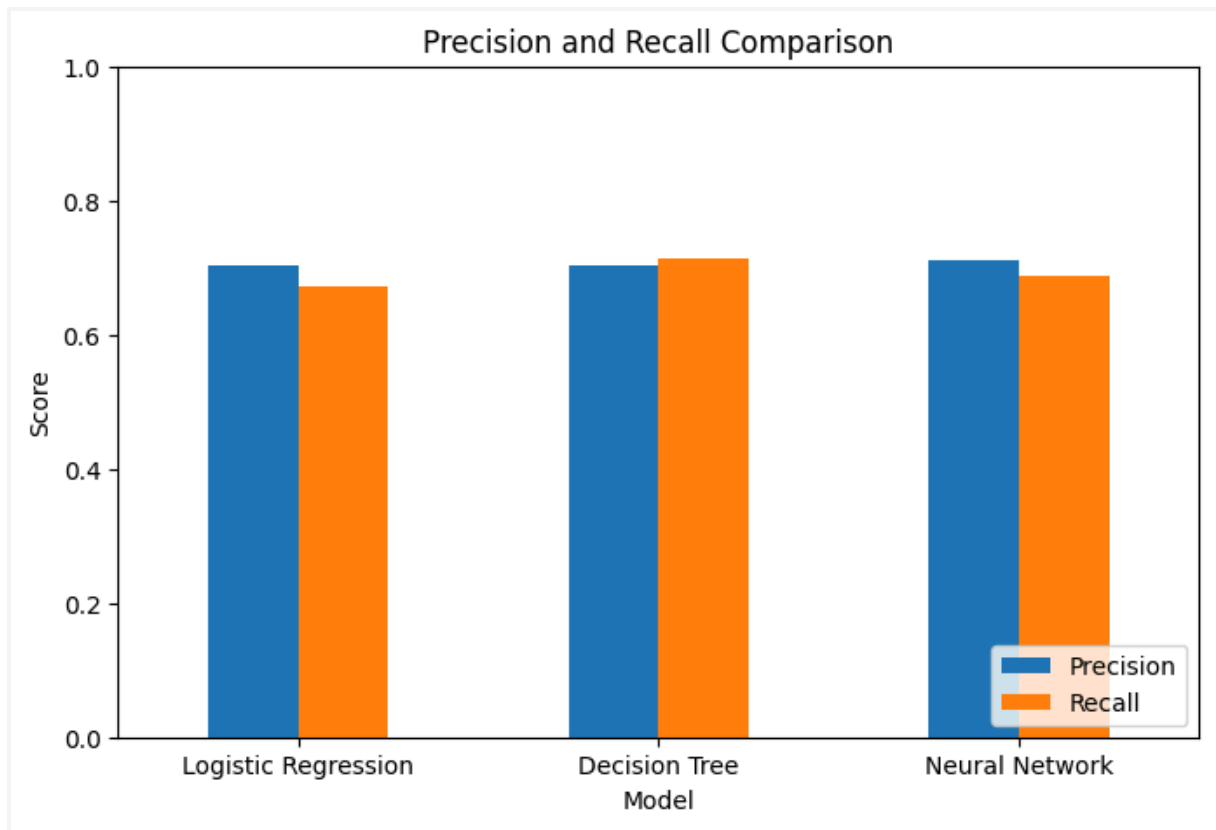
Model selection and comparison analysis

In this project, multiple models including Logistic Regression, Decision Tree, Neural Network, and K-Means clustering were implemented and compared to evaluate their effectiveness in predicting delivery delays. The supervised models were assessed

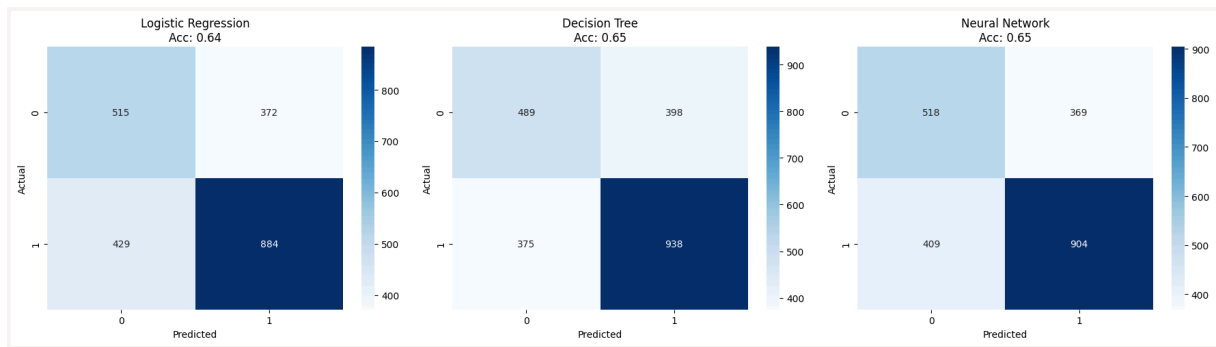
using accuracy, precision, recall, confusion matrix, and ROC-AUC score, while K-Means was included to analyze unsupervised learning performance.



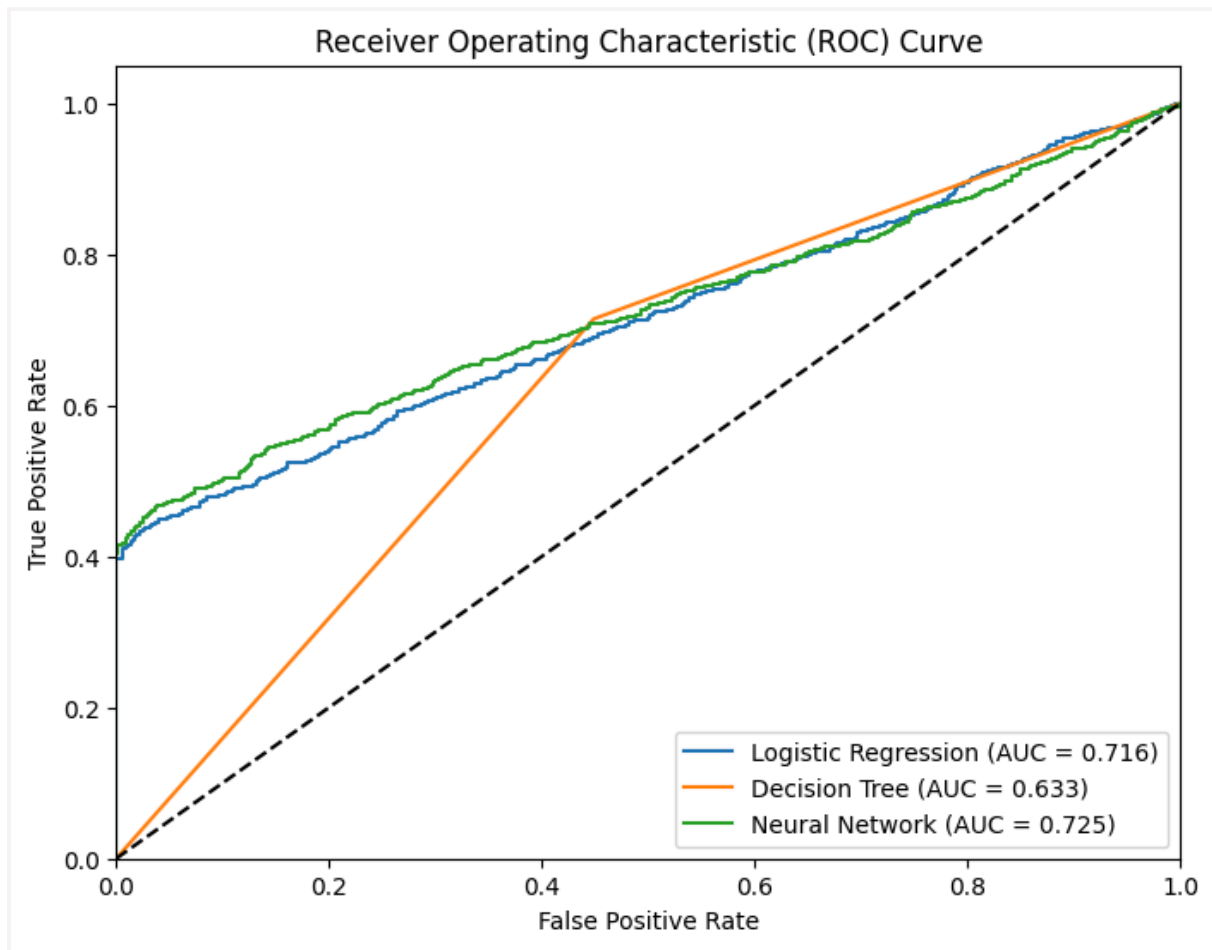
The model accuracy comparison indicates that all three supervised models achieved similar overall performance, with accuracies ranging between 63% and 65%. The Decision Tree produced the highest accuracy at approximately **64.86%**, slightly outperforming the Neural Network, which achieved around **64.64%**. Logistic Regression obtained the lowest accuracy at approximately **63.59%**. Although the differences are relatively small, the Decision Tree demonstrated the best overall classification performance among the evaluated supervised models.



The precision and recall comparison shows that all three supervised models achieved relatively similar performance levels. Logistic Regression obtained a precision of approximately 0.70 and a recall of around 0.67, indicating moderate ability in both correct positive prediction and detection of actual positives. The Decision Tree achieved the highest recall at approximately 0.71, showing that it was the most effective model in identifying actual class 1 deliveries, although its precision remained around 0.70. The Neural Network demonstrated the highest precision at approximately 0.71 while maintaining a recall close to 0.69, providing a balanced trade-off between minimizing false positives and correctly detecting positive cases. Overall, the Decision Tree prioritized recall performance, while the Neural Network offered slightly stronger precision and balanced classification capability.



The confusion matrices show that all supervised models successfully identified a large number of on-time deliveries (class 1), represented by high true positive (TP) values. For Logistic Regression, the model produced 884 true positives and 515 true negatives (TN), while generating 372 false positives (FP) and 429 false negatives (FN). The Decision Tree achieved the highest true positive count with 938 TP and 489 TN, along with 398 FP and 375 FN, indicating strong recall performance for class 1 prediction. The Neural Network demonstrated a balanced performance with 904 TP and 518 TN, while producing 369 FP and 409 FN, giving a comparatively stable trade-off between correctly identifying both classes. Overall, the Decision Tree showed the strongest ability to detect class 1 instances, while the Neural Network maintained slightly better balance across true positives and true negatives. K-Means clustering, in comparison, showed weaker separation capability and higher misclassification behavior, suggesting that the dataset does not naturally form clearly separable clusters without supervision.



The ROC curve analysis further highlights the comparative performance of the supervised models in distinguishing between the two classes across different classification thresholds. The Neural Network achieved the highest AUC score of approximately 0.725, indicating the strongest overall class separation capability among the evaluated models. Logistic Regression followed closely with an AUC of around 0.716, demonstrating similarly strong discriminatory performance despite its simpler linear structure. The Decision Tree obtained a comparatively lower AUC of approximately 0.633, suggesting that although it achieved slightly higher accuracy and recall, its overall ability to consistently separate the classes across varying thresholds was weaker. Overall, the ROC and AUC results indicate that the Neural Network and Logistic Regression provided more reliable threshold-independent classification performance.

Conclusion

From the experimental results, it can be observed that all three supervised learning models — Logistic Regression, Decision Tree, and Neural Network — achieved relatively similar performance, with accuracies around 64–65%. Among them, the Decision Tree produced the highest overall accuracy and recall, indicating a stronger ability to correctly identify class 1 deliveries. The Neural Network demonstrated a more balanced performance between precision and recall, while Logistic Regression provided comparatively lower but stable results as a baseline linear model. The confusion matrices further showed that all models were able to correctly classify a large number of deliveries, although false positives and false negatives were still significant, indicating that the dataset contains overlapping patterns and is not perfectly separable.

The results suggest that delivery prediction is influenced by multiple interacting factors rather than a single dominant feature. Although features such as `Discount_offered` showed strong correlation with delayed deliveries, the PCA and K-Means clustering analysis indicated that the data does not naturally form clearly separable groups. This explains why even more advanced models like Neural Networks did not achieve dramatically higher accuracy than simpler models. The supervised models performed better because they learned directly from labeled data, whereas K-Means struggled due to the absence of label information and overlapping feature distributions.

Several challenges were encountered during the project. One challenge was selecting appropriate preprocessing techniques, particularly for categorical feature encoding and feature scaling. Another challenge was handling overlapping class distributions, which limited the ability of both supervised and unsupervised methods to perfectly separate the classes. Additionally, evaluating K-Means alongside supervised models required special handling because unsupervised clustering does not naturally provide class labels or probability scores. Despite these challenges, the project successfully demonstrated the application and comparison of multiple machine learning techniques for classification and clustering analysis.